

Institute of Actuaries of India

Subject CS1-Actuarial Statistics (Paper A)

March 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) The distribution of the number of claims, N, in one year, is Poi(0.3). Hence the probability of no claims in one year is

$$P(N=0) = \frac{0.3^0}{0!} e^{-0.3} = 0.740818 \quad (1)$$

- ii) The probability of one or more claims in one year is

$$P(N \geq 1) = 1 - P(N=0) = 1 - 0.740818 = 0.259182$$

If X is the number of years with one or more claim, then

$$X \sim \text{Bin}(4, 0.259182)$$

So we have:

$$P(X=2) = {}^4C_2 * 0.259182^2 * (1-0.259182)^2 = 0.221199 \quad (2)$$

- iii) The waiting time, T, in years follows Exp(0.3) distribution

$$P(T > 3) = 1 - F[3] = e^{-0.3*3} = 0.40657 \quad (2)$$

- iv) To simulate a value from a discrete distribution, the following steps would be followed

1. Calculate DF, F(n)
 2. For a random number r between 0 and 1, if $F(n-1) < r \leq F(n)$, then the simulated value is n
- (1)

- v) The CDF is

N	0	1	2
P(N ≤ n)	0.74	0.96	0.99

Since $0.74 < 0.7521 \leq 0.96$, the first simulated value is 1. Since $0 < 0.6521 < 0.74$ the second simulated value is 0

(2)

[8 Marks]

Solution 2:

- i) Sample variance = $\frac{1}{n-1} (\sum x_i^2 - n\bar{x}^2)$

Sample	Sample Variance
1	6.385
2	22.814

(1)

- ii)

- a) Test Statistic : $\frac{\bar{X} - \mu}{\sqrt{S^2/n}} \sim t_{n-1}$

Sample 1

$$t = (19.8 - 18) / \sqrt{6.385/15} = 2.758$$

$$P\text{-value} = 2 * P(t_{14} > 2.68) = 0.0153 < .02$$

We reject null hypothesis and accept $H_1: \mu \neq 18$ at 98% level of confidence (2)

- b) Sample 2

Confidence interval is given by

$$\bar{X} \mp t_{0.01,14} * \sqrt{s^2/n}$$

$$19.8 \mp 2.6245 * \text{sqrt}(22.814/15) = (16.56, 23.04)$$

Since 18 lies between the confidence interval, we cannot reject null hypothesis at 98% level of confidence (2)

c) Sample 2 does not provide enough evidence to justify rejecting H₀, despite having the same size and mean to Sample 1.

The reason for the loss of significance is the much greater variation in the data in Sample 2 – the variance is three times bigger than in Sample 1 (22.814 v 6.386)

– this greatly increases the standard error of estimation. (2)

d) With the larger sample of 100 claims the standard error of the sample mean will be smaller, giving a narrower confidence interval. (2)

e) The replacement of the extreme value will lead to

- [Bonus] Sample mean, which means that the interval will be shifted to the left.
- Reduction of variance of the sample will also be smaller with replacement of outlier value,
- which will again give a narrower interval.

(2)

[11 Marks]

Solution 3:

i) Option C (1)

$$X \sim \text{Gamma}(5, 1/8)$$

$$\text{From Tables, } M_X(t) = (1 - \frac{t}{\lambda})^{-\alpha} = (1 - 8t)^{-5}$$

$$\text{ii) } Y = \frac{1}{4}X, M_Y(t) = E[e^{tY}] = E[e^{(1/4)tX}] = M_X(t/4) = (1 - 8(t/4))^{-5} = (1 - 2t)^{-5}$$

So the moment generating function of Y is $(1 - 2t)^{-5}$

By comparing this with the MGF of the Gamma distribution in the tables, we see that this is the same as the MGF of Gamma(5, 1/2) distribution. Looking at the definition of the chi-square distribution, we see that Gamma(5, 1/2) is equivalent to chi-square distribution with 10 degrees of freedom

By the uniqueness property of moment generating function, therefore we have shown that $\frac{1}{4}X \sim \chi_{10}^2$ (2)

iii) $X \sim \text{Gamma}(5, 1/8)$ where X is the claim amount in units of 1000 INR

$$2\lambda X \sim \chi_{2\alpha}^2 \text{ i.e. } (1/4)X \sim \chi_{10}^2$$

$$P(X > 40) = (X/4 > 10) = P(\chi_{10}^2 > 10) = 1 - 0.5595 = 0.4405 \quad (2)$$

(iv) Since X and Y are independent. The MGF of X + Y is given by the product of MGFs.

$$M_{X+Y}(t) = E[e^{tX}] E[e^{tY}] = (1 - 8t)^{-5} * (1 - 4t)^{-3}$$

$$M_Z(t) = (1 - 8t/3)^{-8}$$

$M_Z(t) \neq M_{X+Y}(t)$, and therefore $X + Y$ does not have a Gamma(8,3/8) distribution (2)
[7 Marks]

Solution 4:

i) Mean and Variances of policies

Policy Number (i)	$\bar{x}_i$	s_i^2
1	4,061	6,252
2	4,366	6,359
3	4,274	6,216
4	4,484	6,121

$E[m(\theta)] = \text{Overall mean} = 4296.25$

$E[s^2(\theta)] = \text{Mean of the variances} = 6237$

$$\begin{aligned} \text{Var}[m(\theta)] &= \frac{1}{3} \sum_{i=1}^4 (\bar{x}_i - \bar{x})^2 - \frac{1}{3} \left[\frac{1}{4} \sum_{i=1}^4 \frac{1}{2} \sum_{j=1}^3 (x_{ij} - \bar{x}_i)^2 \right] \\ &= \frac{1}{3} [(4061 - 4296.25)^2 + (4366 - 4296.25)^2 + (4274 - 4296.25)^2 + (4484 - 4296.25)^2] - \frac{1}{3} \times 6237 \\ &= 29,905 \end{aligned}$$

The estimated credibility factor is $Z = \frac{3}{3 + \frac{6237}{29905}} = 0.935$

Thus, the credibility premium for Policy Number 4 is
 $= 0.935 \times 4,484 + 0.065 \times 4296.25 = 4,471.8$

(4)

ii) EBCT Model 2 can be used with risk volumes $P(i,j)$ for each policy number for each year.
 $P(i,1) = 1$, $P(i,2) = 1/1.2$ and $P(i,3) = 1/1.2^2$

(Alternate solution: Adjust year-wise figures to bring it year 3 level. Multiply 1.2^2 to year 1 and 1.2 to year 2. And then apply the EBCT Model 1 to adjusted on-level figures.)

Data Required: Year wise data will be required since only totals are given. (3)

iii) Policy number 1 has a lowest mean. Removing it will reduce the variance of means, $\text{var}[m(\theta)]$.
 The variance of policy number is similar to other policies and close to overall mean of variance.

Thus, $E[s^2(\theta)]$ will remain similar and won't change much.

Hence, proportionately smaller $\text{var}[m(\theta)]$ will lead to reduction of credibility factor.

(2)

[9 Marks]

Solution 5:

i) The integral over the domain

$$\iint_0^\infty f(x,y) dx dy = 1$$

$$\iint_0^\infty f(x,y) dx dy = k \iint_0^\infty e^{-(3x+\frac{y}{5})} dx dy = k \int_0^\infty e^{-3x} dx \int_0^\infty e^{-\frac{y}{5}} dy$$

$$\int_0^{\infty} e^{-3x} dx = -\frac{e^{-3x}}{3} \Big|_0^{\infty} = \frac{1}{3}$$

$$\int_0^{\infty} e^{-\frac{y}{5}} dy = -5 e^{-\frac{y}{5}} \Big|_0^{\infty} = 5$$

$$\text{Hence } k \cdot \frac{1}{3} \cdot 5 = 1 \text{ Hence } k = 3/5$$

(3)

ii) Answer B

(2)

The marginal distribution of X is

$$f_X(x) = 3/5 \int_0^{\infty} e^{-(3x+\frac{y}{5})} dy = 3/5 * e^{-3x} * \int_0^{\infty} e^{-\frac{y}{5}} dy = 3e^{-3x}$$

The marginal distribution of Y is

$$f_Y(y) = 3/5 \int_0^{\infty} e^{-(3x+\frac{y}{5})} dx = 3/5 * e^{-\frac{y}{5}} * \int_0^{\infty} e^{-3x} dx = 1/5 * e^{-\frac{y}{5}}$$

iii) From the results of point (ii), it is found that

$$f(x, y) = f_X(x) * f_Y(y)$$

Hence joint probability function is the product of two marginal probability functions for all (x,y) in the range of the variables hence X and Y are independent

(1)

iv) The conditional probability $P(X \leq x | X > 5)$ is

$$P(X \leq x | X > 5) = \frac{P(5 \leq X \leq x)}{P(X > 5)} = \frac{F_X(x) - F_X(5)}{P(X > 5)}, x > 5$$

Therefore,

$$f(x | x > 5) = \frac{f_X(x)}{P(X > 5)} = \frac{3e^{-3x}}{e^{-15}} = 3e^{15-3x}, x > 5$$

$$\text{Since } P(X > 5) = \int_5^{\infty} 3e^{-3x} dx = -e^{-3x} \Big|_5^{\infty} = e^{-15}$$

(3)

v) Option D

(1)

The conditional expectation

$$E[X | X > 5] = \int_5^{\infty} x f(x | x > 5) dx = \int_5^{\infty} x 3e^{15-3x} dx$$

By taking $t = x - 5$

$$E[X | X > 5] = \int_0^{\infty} (t + 5) 3e^{-3t} dt = \int_0^{\infty} 3t e^{-3t} dt + \int_0^{\infty} 15 e^{-3t} dt$$

[10 Marks]**Solution 6:**Credibility factor $Z = n / (n + s^2 / \sigma^2)$

No requirement to specify credibility factor.

 μ_0 : No relationship

1 Mark

σ^2 : Z will be an increasing function of σ^2 . Higher variability in prior means less relevance and high spread making the prior less credible for estimation and hence, more credit to sample.

s^2 : Z will be an decreasing function of s^2 . High sample variance indicates high spread of sample values over a wide range and hence, less reliable for estimation. Thus, with increase in sample variance less credibility given to sample value. **[5 Marks]**

Solution 7:

i) Option C (3)

The distribution of sample means are

$$\bar{X} \sim N(500, 10) \quad \bar{Y} \sim N(510, 5)$$

$$\bar{Y} - \bar{X} \sim N(510-500, 5+10) \sim N(10, 15)$$

$$P(\bar{X} > \bar{Y}) = P(\bar{Y} - \bar{X} < 0) = P(Z < \frac{0-10}{\sqrt{15}}) = P(Z < -2.58) = 1 - P(Z \leq 2.58) = 1 - 0.99506 = 0.00494$$

$$\begin{aligned} \text{ii) } P(S_X^2 > 100) &= 1 - P(S_X^2 \leq 100) \\ &= 1 - P\left(\frac{S_X^2}{100} \leq 9\right) \\ &= 1 - P(\chi_9^2 \leq 9) \\ &= 1 - 0.5627 = 0.4373 \end{aligned} \quad (2)$$

iii) Option B (3)

$$\begin{aligned} S_X^2 \text{ and } S_Y^2 \text{ are independent and therefore } \frac{S_X^2/100}{S_Y^2/25} &\sim F_{9,4} \\ P[S_X^2 < S_Y^2] &= P\left[\frac{S_X^2}{S_Y^2} < 1\right] = P\left[\frac{S_X^2/100}{S_Y^2/25} < \frac{25}{100}\right] = P[F_{9,4} < .25] \end{aligned}$$

$$P[F_{9,4} < .25] = P[F_{4,9} > 1/.25] = P[F_{4,9} > 4]$$

The value is between 2.5% and 5%, and interpolating, we find the probability is approximately 4.2%

$$\text{iv) } P(S_X > 4S_Y) = P(S_X/S_Y > 4) = P(S_X^2/S_Y^2 > 16)$$

$$P(S_X^2/S_Y^2 > 16) = P\left(\frac{S_X^2/S_Y^2}{4} > 4\right) = P(F_{9,4} > 4)$$

From Table the required approximate probability is 10% **(2)**
[10 Marks]

Solution 8:

- i) If, taking a sample from the distribution parameterised by λ , the posterior distribution of λ belongs to the same family as the prior distribution then the prior is called a conjugate prior. (2)

- ii) Assume Prior distribution of λ is $\text{Gamma}(\alpha, \beta)$.

$\underline{X}$ is the sample taken from exponential distribution then posterior density satisfies:

$$\begin{aligned} f(\lambda|\underline{X}) &\propto f(\underline{X}|\lambda)f(\lambda) \\ &= \left[\prod_{i=1}^n \lambda e^{-\lambda x_i} \right] \times \frac{\beta^\alpha \lambda^{\alpha-1} e^{-\beta\lambda}}{\Gamma(\alpha)} \\ &\propto \lambda^{\alpha+n-1} e^{-\lambda(\beta + \sum_{i=1}^n x_i)} \\ &\propto \text{pdf of } \Gamma(\alpha+n, \beta + \sum_{i=1}^n x_i) \end{aligned}$$

This means that the posterior distribution of λ also follows a Gamma distribution and therefore the Gamma distribution satisfies the definition of a conjugate prior. (3)

- iii)

- a) Given $\lambda \sim \text{Gamma}(\alpha, \beta)$

$$\begin{aligned} E\left(\frac{n}{\lambda}\right) &= \int_0^\infty \frac{n f(\lambda)}{\lambda} d\lambda \\ &= \int_0^\infty \frac{n \lambda^{\alpha-1} e^{-\beta\lambda}}{\lambda \Gamma(\alpha)} d\lambda \\ &= \frac{n\beta}{\alpha-1} \int_0^\infty \frac{\beta^{\alpha-1} \lambda^{\alpha-2} e^{-\beta\lambda}}{\Gamma(\alpha-1)} d\lambda \\ &= \frac{n\beta}{\alpha-1} \int_0^\infty \text{pdf of gamma}(\alpha-1, \beta) d\lambda \\ &= \frac{n\beta}{\alpha-1} \times 1 \\ &= \frac{n\beta}{\alpha-1} \end{aligned} \quad (2)$$

- b) Using (ii), Posterior distribution of λ is $\text{Gamma}(\alpha+n, \beta + \sum_{i=1}^n x_i)$.

Using (iii a), Posterior mean of n/λ

$$\begin{aligned} &= \frac{n(\beta + \sum_{i=1}^n x_i)}{\alpha+n-1} \\ &= \frac{n\beta}{(\alpha+n-1)} + \frac{n \sum_{i=1}^n x_i}{(\alpha+n-1)} \\ &= \frac{\alpha-1}{(\alpha+n-1)} \times \frac{n\beta}{\alpha-1} + \frac{n}{(\alpha+n-1)} \times \sum_{i=1}^n x_i \end{aligned}$$

$$\text{Consider } Z = \frac{n}{(\alpha+n-1)}$$

$$\text{Using (iii a), } \frac{n\beta}{\alpha-1} \text{ is } E\left(\frac{n}{\lambda}\right)$$

$$\text{Thus, posterior mean of } n/\lambda = Z^* \sum_{i=1}^n x_i + (1-Z) * \text{prior mean of } n/\lambda$$

(3)

[10 Marks]

Solution 9:

- i)

- a) Correct Option is Option D (2)

steps not required

$$\text{theta} = \log(\mu) \text{ hence, } \mu = e^{\text{theta}}$$

$$E(Y) = b(\text{theta}) = \mu = e^{\text{theta}}$$

$$b'(\theta) = e^{\theta} = \mu$$

$$V(\mu) = b''(\theta) = e^{\theta} = \mu$$

Hence correct option is Option D

b) Correct option is Option C (2)

steps not required

$$b(\theta) = -\log(-\theta)$$

$$\text{mean} = E(Y) = b'(\theta) = -1/\theta$$

$$\text{as } \theta = 1/\mu, b'(\theta) = \mu$$

$$\text{variance function is } V(\mu) = b''(\theta) = 1/\theta^2 = \mu^2$$

$$\text{Hence, variance is } V(\mu) / a(\phi) = \mu^2/1 = \mu^2$$

Hence, correct option is Option C i.e. mean = μ and variance = μ^2

ii)

a) **Interaction** term means that effect of age band on accidental hospitalisation claims depends on the gender of the insured and **significant** indicates that accidental hospitalisation claims are better modelled with interaction term, when claims for any age group change with respect to gender of the insured.

E.g. accidental hospitalisation claims are expected to be higher for males compared to females for any age group.

This can be achieved in the model by having $\text{Beta_males} > \text{Beta_females}$ across age groups. This will result in higher expected accidental hospitalisation claims for males compared to females for any age group (assuming only other parameter is for age band which is same for males and females). (4)

b) Policy renewal is binary event for a single policy. Hence, predicted policy renewal rate (say, μ) for a group of policies can vary between 0 to 1.

If logit link function is used, then

$$\eta = \log(\mu/(1-\mu))$$

$$\mu/(1-\mu) = \exp(\eta)$$

$$\mu = \exp(\eta) - \mu * \exp(\eta)$$

$$\mu(1 + \exp(\eta)) = \exp(\eta)$$

$$\mu = \exp(\eta) / (1 + \exp(\eta)) = 1 / (1 + \exp(-\eta)) = (1 + \exp(-\eta))^{-1}$$

This is expected to result in the range of 0 to 1 for μ as required - renewal rate for a group of policies.

Hence, logit link function can be used for renewal rate. (4)

iii)

a)

	Model 1	Model 2	
SSREG	1.380	2.380	Given
A	1.380	2.380	as $A/1 = \text{SS}_{\text{REG}}$
B	4.000	3.000	$B = 5.38 - A$
C	0.333	0.250	$C = B/12$
F	4.140	9.520	$F = \text{SS}_{\text{REG}}/C$

(4)

b)

H_0 = Beta parameter is zero (no linear relationship)

H_1 = Beta parameter is not equal to zero (linear relationship is present)

For Model 1, $F = 4.14$, this is between 5% critical value of 4.747 and 10% critical value of 3.177

Hence, we have sufficient evidence to reject null hypothesis that beta parameter is zero at 10% level (but cannot be rejected at 5% level) indicating linear relationship between response variable and predictor variable at 10% level

For Model 2, $F=9.52$ is greater than critical value at 1% level as well.

Hence, we have sufficient evidence to reject null hypothesis even at 1% level indicating linear relationship between response variable and predictor variable at 1% level

(3)

c) $R^2 = SS_{\text{REG}} / SS_{\text{TOT}}$

For Model 1, $R^2 = 1.38/5.38 = 25.7\%$ and

For Model 2, $R^2 = 2.38/5.38 = 44.2\%$

As % of variation explained by model 1 and Model 2 is low (based on low value of R^2 of Model 1 and Model 2), we can conclude that none of the model is good fit to the data and hence, models are not suitable for prediction purpose (though there is linear relationship between response and predictor variable for Model 1 and Model 2 as indicated in part b)

(3)

iv)

a)

Based on Result 1

- Model using x as predictor is significant improvement over null model
- As reduction in deviance (by 1769) is significantly more compared to 2 times the loss in degrees of freedom by 1 when x is used as predictor over null model
- Result is significant even at 0.000001 level as p value is smaller than that

Based on Result 2

- Model using interaction term between x and region is significant improvement over model using just x as predictor
- As reduction in deviance (by 87) is significantly more compared to 2 times the loss in degrees of freedom by 2 when interaction between x and region is considered over model using just x as predictor
- Result is significant at 0.005 level as p value is smaller than 0.005

(4)

b)

Based on Result 3

- Model using only main effect of x and region is not significantly different compared to model using interaction between x and region
- As reduction in deviance (by 0.037) is less than 2 times the loss in degrees of freedom by 1 when interaction between x and region is considered over model using just the main effect of x and region
- As p value (0.92) is much more than 90% and

Comparing Result 2 and Result 3

- Model using only main effects of x and region is significantly better than model using x as predictor

- as reduction in deviance is significantly more compared to 2 times the loss in degrees of freedom by 1 when main effect of x and region are used as predictor over model using only x
- and p value to be significant at 0.005 level
- We prefer simpler model i.e. Model with only main effects of x and region over complex model i.e. Model having interaction between x and region as additional complexity is not justified by better prediction

(4)

[30 Marks]
